Computer Technology Competence of IT Faculty in Non-Accredited State Universities and Colleges in Region VI: A Focused Secondary Analysis

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ABSTRACT

This paper presents a focused analysis of the computer-technology competence findings reported in a study of 154 Information Technology faculty members from non-accredited IT programs of state universities and colleges in Region VI. The original study used a revised standardized competence instrument covering computer operation, setup and troubleshooting, word processing and desktop publishing, spreadsheet and graphing, database, networking, telecommunications, and media communications. The overall competence mean was 4.5508 (SD = .4490), interpreted in the source as Very High. All demographic and institutional subgroups were likewise classified as Very High. Inferential tests found no significant competence differences by age, sex, educational qualification, teaching experience, or school location, but a significant difference by academic rank (Kruskal-Wallis chi-square = 6.697, p = .035). The paper discusses these findings as evidence of broadly distributed perceived technology competence while identifying academic rank as the only grouping variable

associated with statistically different competence distributions in the source analysis.

Keywords: *computer technology competence, IT faculty, state universities and colleges, academic rank, Region VI, accreditation*

INTRODUCTION

Faculty competence was one of the central constructs in Lagon's study of non-accredited Information Technology programs in state universities and colleges in Western Visayas. The dissertation treated computer-technology competence as a major consideration in accreditation readiness and used a standardized instrument adapted from Saud (2005). The instrument contained eight sections spanning basic computer operation through networking, telecommunications, and media communications.

The present manuscript narrows the original dissertation to one question: how did the IT faculty members assess their computer-technology competence, and which faculty characteristics were associated with differences in competence? No new respondents or raw-data reanalysis are introduced; all values come from the published aggregate tables.

Methodological Source

The original study involved 154 part-time and permanent IT faculty members from non-accredited IT programs of SUCs in Region VI. Respondents were grouped by age, sex, educational qualification, length of teaching experience, academic rank, and school location. Means and standard deviations described competence, t-tests were used for two-category groupings, and the Kruskal-Wallis test was used for academic rank and school location at a .05 significance level.

Descriptive Competence Profile

Grouping Category Mean SD Source interpretation

Entire group 154 respondents 4.5508 .4490 Very High
Age 30 and below 4.5198 .4221 Very High
Age Above 30 4.5870 .4791 Very High
Sex Male 4.5871 .4242 Very High
Sex Female 4.4907 .4852 Very High
Education Master's degree 4.6438 .3363 Very High
Education Baccalaureate degree 4.5147 .4801 Very High
Teaching experience Above 8 years 4.6018 .5647 Very High
Teaching experience 8 years and below 4.5329 .4022 Very High
Academic rank Part-time instructor 4.4821 .4251 Very High
Academic rank Instructor 4.6079 .4161 Very High
Academic rank Professor 4.5862 .6915 Very High

The most important descriptive feature is the narrow interpretive range: every reported subgroup falls within the source's Very High category. Master's-degree holders, faculty with longer teaching experience, instructors, and older faculty generally have

Five derivative journal manuscripts based on Lagon's published dissertation Page 3

somewhat higher means than their comparison groups, but descriptive mean differences alone do not establish statistical significance.

Which Group Differences Were Statistically Significant?

Grouping variable Test statistic p value Decision in source
Age t = -0.926 .356 Not significant
Sex t = 1.294 .198 Not significant
Educational qualification t = -1.568 .119 Not significant
Length of teaching experience t = -0.834 .406 Not significant
Academic rank Kruskal-Wallis chi-square = 6.697 .035 Significant
School location Kruskal-Wallis chi-square = 10.045 .074 Not significant

Academic rank is the only grouping variable for which the source rejects the null hypothesis. Mean ranks reported for part-time instructors, instructors, and professors were 67.59, 82.97, and 95.50, respectively. This pattern suggests that competence distributions differed across ranks even though all three rank groups remained in the same Very High descriptive category.

The distinction between descriptive level and inferential difference is important. A group can be classified as Very High while still differing significantly from another Very High group. Conversely, slightly different means do not necessarily imply a statistically reliable difference.

Implications for Faculty Development

The source recommendation that SUCs maintain and uphold computer-technology competence is consistent with these results. Because the competence level was broadly high across sex, age, education, experience, and location, a universal basic remediation program may be less appropriate than differentiated professional development focused on advanced and role-specific competencies.

The academic-rank result also supports attention to career development. The dissertation recommends scholarships and financial assistance that may help younger and part-time instructors pursue graduate study and strengthen faculty profiles. Such recommendations should be viewed as institutional-development strategies derived from the observed rank pattern, not as causal effects demonstrated by this cross-sectional study.

LIMITATIONS

The competence instrument is perception-based, and the published dissertation provides aggregate rather than item-level or performance-test data. Therefore, the results describe reported competence within the source framework. They do not establish direct proficiency through an external skills examination, nor do they show how competence changed over time.

CONCLUSION

IT faculty in the studied non-accredited SUC programs reported very high computer-technology competence across all examined subgroups. Academic rank was the only grouping variable associated with a significant difference in competence distributions, while age, sex, educational qualification, teaching experience, and location were not. For institutional planning, the findings support maintaining high baseline competence while targeting advanced development and career pathways rather than assuming broad technological deficiency.

REFERENCE

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